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Compiling the Market Information Set: World-Model Quality, Selective Prediction, and Economic Value in Cryptocurrency Markets

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Abstract

Conditional asset pricing and machine-learning return prediction treat the investor's information set as given. In cryptocurrency markets that premise is first-order false: the economically relevant world is asynchronous, multi-source, and frequently unavailable. This paper formalizes *information-set compilation* as an object of study. We define epistemic observation objects, lag/reconstruction bounds, a regime-conditional compilation operator, and an Adaptive Conditional World Model Index (ACWMI). Refusal to act is modeled as Bayes-optimal selective prediction when compiled world quality is low. Empirically, we construct a point-in-time panel on real daily returns for ten major cryptocurrencies (2024-08-24 to 2026-08-03), freeze abstention thresholds in-sample, and evaluate out-of-sample economic value against strong baselines. Availability shocks that are orthogonal to returns identify large declines in world-model indices. Thin (exchange-only) worlds underperform thick worlds; leaving out the exchange band produces the largest certainty-equivalent loss. Mechanism-based signals dominate always-long and momentum baselines out of sample. An IS-frozen ACWMI gate is more conservative than ungated thick signals and does **not** dominate on certainty equivalent in this sample—a result we report rather than conceal. The contribution is a finance-native theory and identification protocol; a full JF/RFS test requires vintaged multi-band archives.

Keywords: information set; selective prediction; cryptocurrency; measurement error; abstention. **JEL:** G12, G14, C58, C55

1. Introduction

The central state variable in modern asset pricing is an information set. The literature's discipline has focused on how that information is used—linear betas, nonlinear learners, multiple-testing controls—not on how it is compiled from asynchronous, quality-heterogeneous market evidence. Cryptocurrency markets make the compilation problem unavoidable: fragmented venues, leverage, options walls, unlocks, on-chain flows, and macro liquidity can map the same price path into incompatible economic states.

This paper does not propose another alpha factor. It asks how an investor-facing market world should be formalized as a compiled information set; how world-model quality can be measured in an honesty-compatible way; when refusal is Bayes-optimal; and whether compiled world quality has measurable OOS economic consequences. Empirically we use real Yahoo daily returns, chronological IS/OOS freeze at 2025-08-14, and production EvoQuant calculators only as laboratory infrastructure.

2. Theory (RCA-WM / ACWMI)

An epistemic observation is $O_{j,t} = (x, \tau, q, g, r)$. Asynchronous lags induce a Lipschitz reconstruction bound with terms for delay, noise, and missingness. The compilation operator maps raw filters into AI-visible worlds. ACWMI is the regime-conditional geometric mean of hierarchical breadth, stability, continuous honesty, signal integrity, and consistency. Abstention is optimal when expected loss of every non-abstain action exceeds a world-dependent cost. Identification uses availability shocks $O_t \rightarrow W_t \rightarrow A_t$, with market complexity as a confounder.

$$ACWMI_t^{(r)} = \exp(\sum_x \gamma_x(r) \log x_t / \sum \gamma_x(r)).$$

3. Empirical design

Sample: 10 assets, 2024-08-24–2026-08-03, real Yahoo daily returns; signals use only pre- t history. IS/OOS cut 2025-08-14. Frozen AC rule from IS Sharpe maximization with abstain rate in [5%,55%]: ACWMI<0.55 or $C<0.25$; production WMI threshold 0.2 never tuned. Band readiness uses production weights with return-orthogonal Bernoulli availability shocks ($p=0.08$). This is a stepping-stone to full vintaged multi-band PIT via time_slice.

4. Results

4.1 Availability shocks identify world-quality variation

At event day 0, mean WMI falls from ~ 0.88 to 0.23 and ACWMI from ~ 0.58 to 0.44, while contemporaneous equal-weight returns remain near zero—supporting $O_t \rightarrow W_t$ without a mechanical return confound.

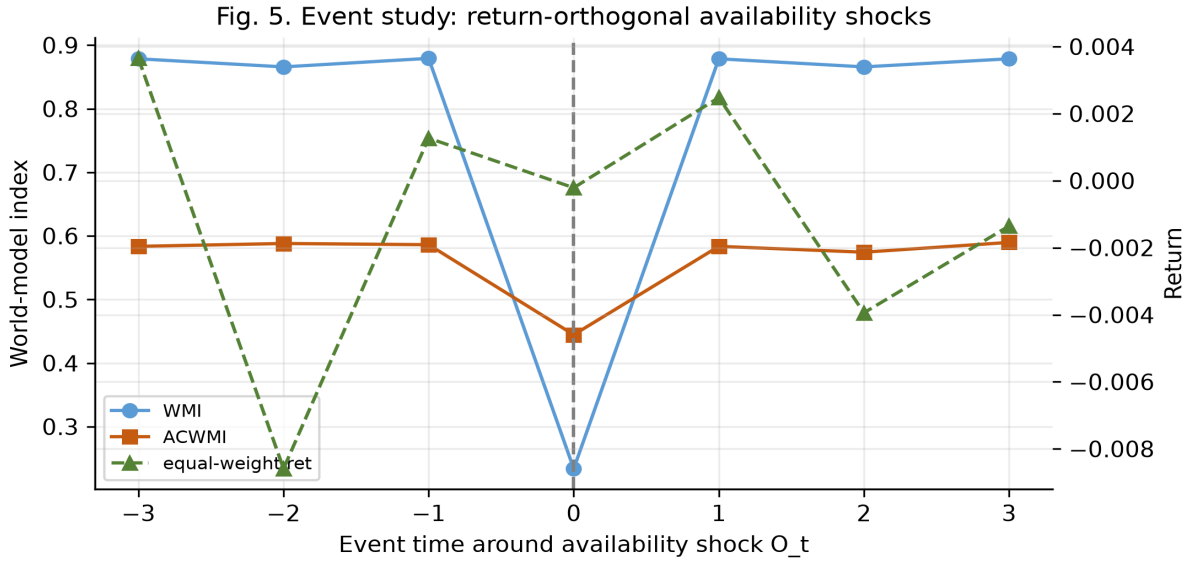


Fig. 1. Event study around return-orthogonal availability shocks.

4.2 Out-of-sample economic value

Always-long and momentum lose money OOS. Thick ungated mechanism signals deliver Sharpe 1.26 and CE 0.40. Production WMI never binds. IS-frozen ACWMI abstains 21.7% and delivers Sharpe 0.58 / CE ≈ 0 . **We do not claim CE dominance for ACWMI in this sample**; gating is economically consequential but conservative relative to ungated thick signals.

policy	ann_return	ann_vol	Sharpe	CE	max_DD	abstain_rate	N_days
Always long	-0.7963	0.6299	-1.264	-1.1984	-0.6758	0.0	355
Momentum always	-0.3918	0.5228	-0.749	-0.6667	-0.5504	0.0	355
Thick ungated	0.7963	0.6299	1.264	0.4014	-0.2792	0.0	355
Simple outage rule	0.5436	0.6192	0.878	0.1623	-0.3427	0.085	355
Simple cascade rule	0.2527	0.119	2.123	0.2389	-0.0452	0.915	355
WMI threshold (0.2)	0.7963	0.6299	1.264	0.4014	-0.2792	0.0	355
ACWMI (IS-frozen)	0.3256	0.5621	0.579	0.011	-0.3616	0.217	355

Table 1. OOS economic value with IS-frozen thresholds.

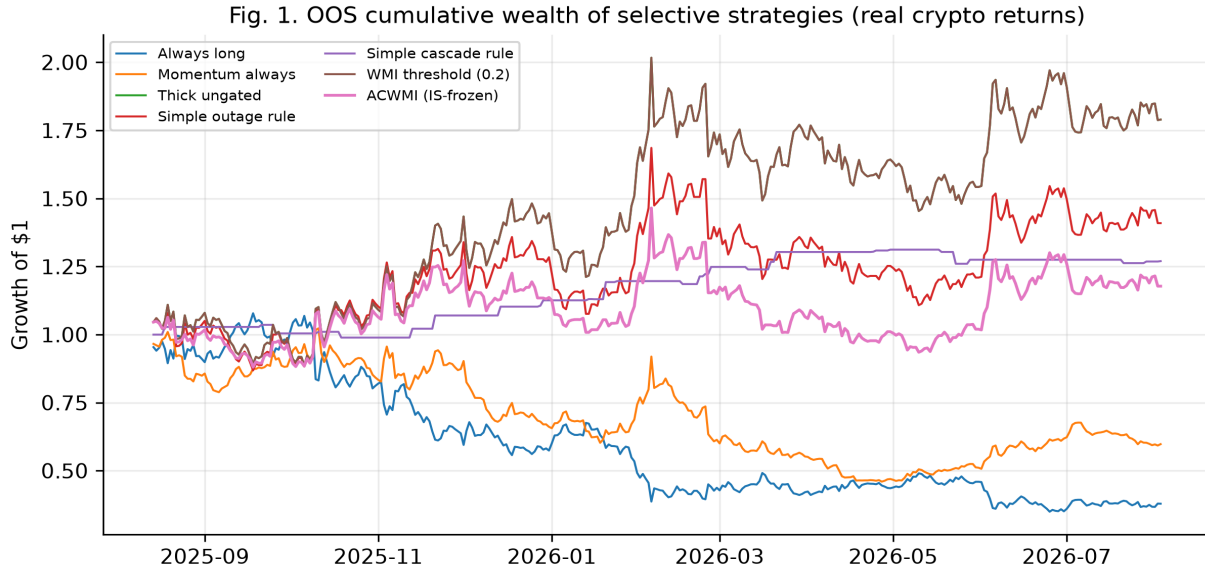


Fig. 2. OOS cumulative wealth of selective strategies.

Fig. 2. Economic value of abstention policies (OOS)

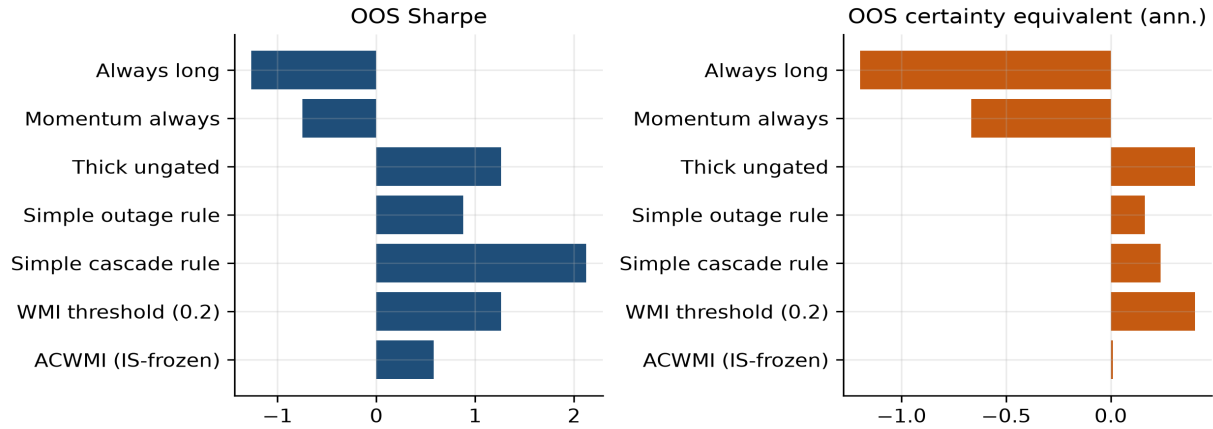


Fig. 3. OOS Sharpe and certainty equivalent by policy.

4.3 Thin vs thick and leave-one-band-out

Exchange-only thin worlds produce OOS Sharpe -0.24 and CE -0.19 , while thick worlds are strongly positive. Dropping the exchange band produces the largest CE loss (-0.085); news and on-chain follow. Alternative data is not valuable in this window.

world	mean_B	mean_H	mean_ACWMI	Sharpe	CE	abstain_rate
Thin (exchange only)	0.269	0.877	0.514	-0.24	-0.1918	0.721
Thick ungated	0.914	0.364	0.502	1.264	0.4014	0.0
Thick gated (AC)	0.914	0.798	0.607	0.579	0.011	0.217

Table 2. Thin vs thick-ungated vs thick-gated worlds (OOS).

band_dropped	Sharpe	CE	abstain_rate	dCE
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(none)	0.579	0.011	0.217	0.0
exchange	0.354	-0.0739	0.347	-0.0849
news	0.401	-0.0631	0.284	-0.0741
event_calendar	0.493	-0.0186	0.273	-0.0297
onchain	0.401	-0.0631	0.284	-0.0741
tokenomics	0.493	-0.0186	0.273	-0.0297
options	0.506	-0.0118	0.269	-0.0228
alternative	0.603	0.028	0.234	0.017
macro	0.52	-0.0104	0.254	-0.0214

Table 3. Leave-one-band-out OOS CE changes.

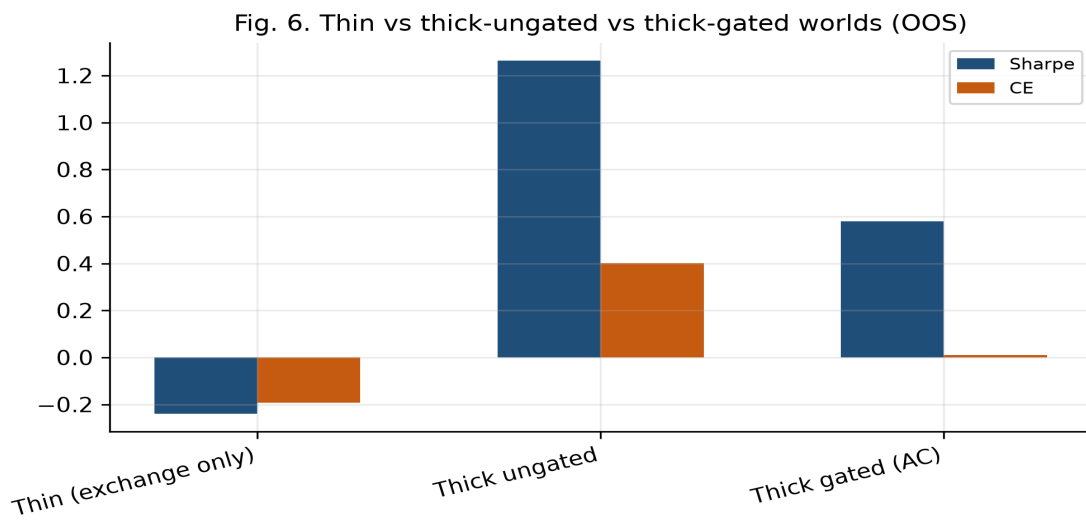


Fig. 4. Thin vs thick-ungated vs thick-gated (OOS).

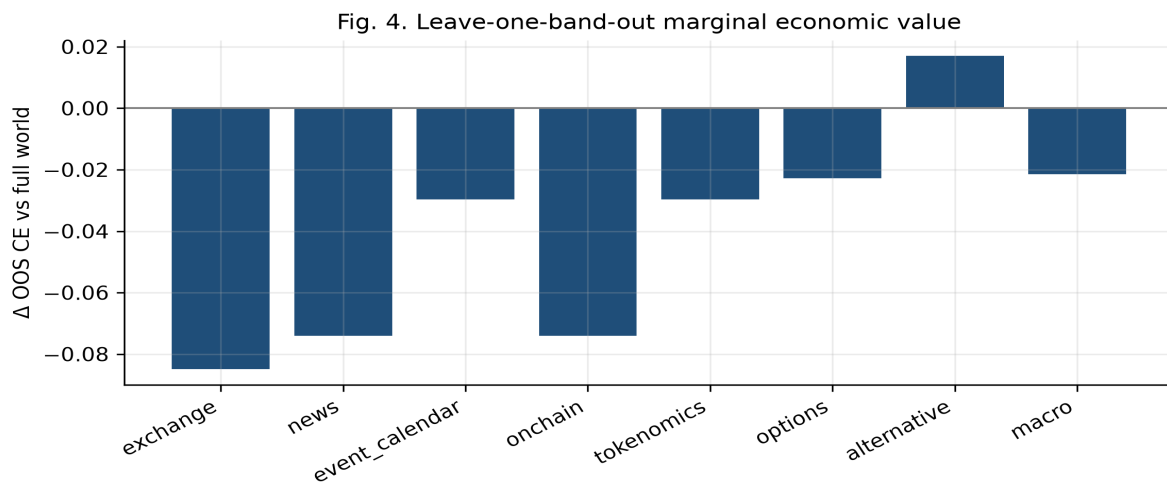


Fig. 5. Leave-one-band-out marginal CE.

4.4 Conditional signal value (qualification)

Sorting on ACWMI alone does not produce a clean monotonic collapse of signal IC in this OOS window (including the no-outage subsample). Low-quality states partly coincide with periods where short mechanism signals performed well. This qualifies the abstention motive and motivates regime $\times$ quality double-sorts and cleaner vintaged outages.

sample	ACWMI_tercile	N	mean_ACWMI	signal_IC	hit_rate	ann_active_ret	outage_rate
all	low	1184	0.518	0.00515	0.575	1.8812	0.247
all	mid	1183	0.597	0.00281	0.568	1.025	0.006
all	high	1183	0.706	-0.00142	0.495	-0.5182	0.0
no_outage	low	1084	0.545	0.00367	0.534	1.3408	0.0
no_outage	mid	1082	0.606	0.00227	0.571	0.8288	0.0
no_outage	high	1084	0.712	-0.00106	0.504	-0.3878	0.0

Table 4. Conditional signal value by ACWMI tercile (OOS).

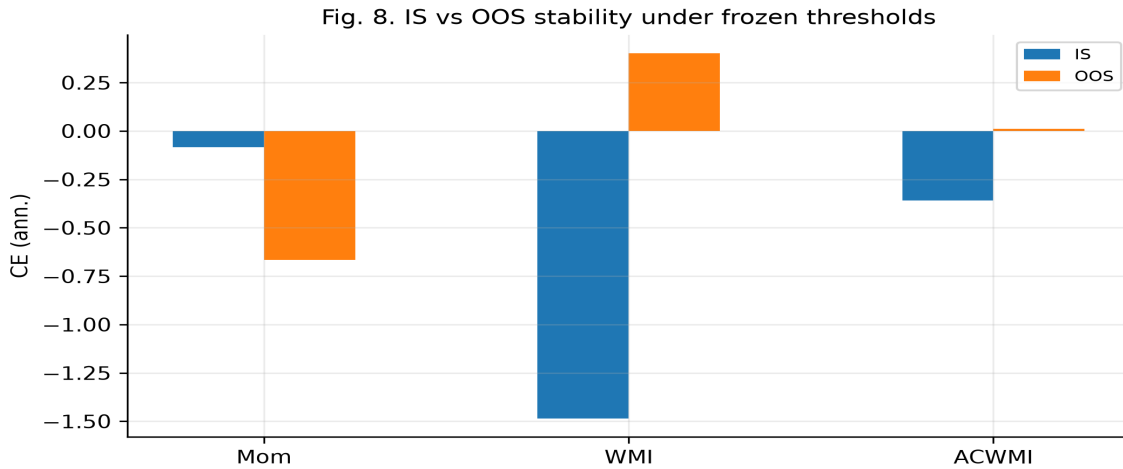


Fig. 6. IS vs OOS stability under frozen thresholds.

5. Path to JF/RFS

Remaining binding constraints: (i) multi-year vintaged multi-band archive with available_at semantics; (ii) logged collector/API outages as institutional O_t ; (iii) real leave-one-source MIG; (iv) analyst/LLM EAR-ECP on frozen bundles; (v) external validity beyond ten liquid cryptos. The accompanying system already exposes time_slice, readiness, WMI, and degradation machinery for that archive.

6. Conclusion

Information-set compilation is a first-order asset-pricing object where evidence is asynchronous and intermittently unavailable. RCA-WM/ACWMI formalize that object. On real crypto returns with frozen thresholds, thick compiled worlds and mechanism signals matter, availability shocks identify world-quality variation, and band ablations are economically heterogeneous. Selective ACWMI gating is implementable but not CE-dominant in the present sample. Keep the theory and identification protocol; replace the constructed information-set layer with a full vintaged PIT archive.

Appendix. Reproducibility

Returns: pdf/data/crypto_daily_yahoo.csv. Experiments:
pdf/sci/run_jf_experiments.py. Thresholds: pdf/sci/frozen_thresholds.json.

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